Do Now · retrieval
OOP debugging with IDEs

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which structure supports BFS traversal?

Q2

What is an object's field?

Q3

When a local variable shadows a field, which name is used without qualification?

Q4

Why might Step Into be more useful than Step Over for a faulty function?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Set a breakpoint — Pause at a method call or the following instruction to inspect current state.

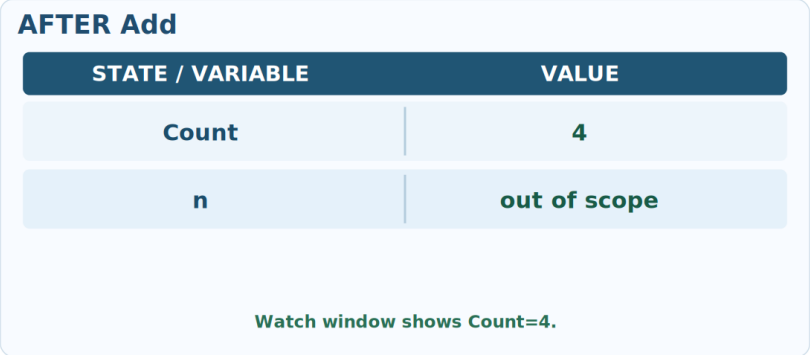
Core idea 2

Step into to inspect internals — Step Into executes the called method one instruction at a time. Watch Count and parameter n.

Core idea 3

Step over to skip internals — Step Over runs the entire call and pauses at the next statement, allowing you to inspect the result.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

breakpoint — A chosen instruction where execution pauses.

Step Into — Enter a called function for debugging.

Step Over — Execute a call without entering it line by line.

watch — IDE display of chosen variables/expressions.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A Score class has Points=3. Add(5) should change Points to 8. Identify a wrong line.

Follow the worked method

1. Predict first — Step Into Add and inspect the local points and the field Points.

2. Model the reasoning — The local becomes 8 while the object field remains 3; the assignment targets the wrong variable.

3. Model the reasoning — Replace it with Points += amount; and rerun. The field should be 8.

Source model · final illustrated step

CORRECTED

STATE / VARIABLE	VALUE
Points field	8
fix	Points += amount

Use the field in the assignment.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

After the first call, true is correct; after the second, IsOn should become false.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain Step Into and Step Over at a method call.

Question 2

A class method Increase(int n) contains `int Total = Total + n;` while Total is an existing property. Identify the issue and write correct C# code.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Design an IDE-debugging workflow for a temperature sensor whose Readings list gains two entries after a single AddReading() call.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Step Over a method call means it is not executed.

Use key vocabulary from memory

1. Explain breakpoint in your own words.

2. Explain Step Into in your own words.

3. Explain Step Over in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.